Week 6 In-Class Worksheet
Power, Two-Sample Tests, and ANOVA

Name: ______________________________ **Date:** ______________

Course: H524 Introduction to Biostatistics
Topic: Statistical Power, Two-Sample Comparisons, and Analysis of Variance

Instructions

Work with a partner to solve the following problems. Show all your work. You may use R to verify your answers and create visualizations, but demonstrate your understanding of the concepts by showing calculations and explaining your reasoning.

1 Problem 1: Statistical Power

A pharmaceutical company is planning a clinical trial to test if a new drug reduces systolic blood pressure. Based on previous studies:

- Population mean blood pressure: 140 mmHg
- Standard deviation: 18 mmHg
- They want to detect a reduction to 135 mmHg (5 mmHg decrease)
- Significance level: $\alpha = 0.05$ (two-sided test)

Part A: If they enroll 50 patients, what is the power of their study? Use R to calculate.

R code:

Power: ____________

Part B: What sample size is needed to achieve 80% power?

R code:

Required n: ____________

Part C: What does "80% power" mean in the context of this study? Write a complete sentence.

Part D: If the company can only afford to enroll 100 patients, what is the smallest effect (reduction in mmHg) they could detect with 80% power?

Hint: Set `n = 100`, `power = 0.80`, and solve for `delta`.

2 Problem 2: Two-Sample t-Test

A nutritionist compares the effectiveness of two weight-loss diets. She randomly assigns 12 participants to Diet A and 12 to Diet B. After 8 weeks, weight loss (in kg) is measured:

Diet A: 3.2, 4.5, 2.8, 5.1, 3.9, 4.2, 3.6, 4.8, 3.3, 4.0, 3.7, 4.4
Diet B: 5.2, 6.1, 4.8, 5.9, 5.5, 6.3, 5.0, 5.7, 5.4, 6.0, 5.3, 5.8

Part A: Calculate summary statistics for each group. Use R.

Diet A: $\bar{x}_A =$ __________, $s_A =$ __________, $n_A =$ ________

Diet B: $\bar{x}_B =$ __________, $s_B =$ __________, $n_B =$ ________

Part B: State the null and alternative hypotheses for testing if Diet B results in greater weight loss than Diet A.

H_0: ____

H_A: ____

Part C: Perform a two-sample t-test in R. Use `var.equal = TRUE` (assume equal variances). Use a one-sided test.

R code:

Test statistic: $t =$ __________

Degrees of freedom: $df =$ __________

p-value: __________

Part D: What is your conclusion at $\alpha = 0.05$? Write a complete sentence in context.

Part E: Calculate and interpret a 95% confidence interval for the difference in mean weight loss ($\mu_B - \mu_A$).

95% CI: ____________________

Interpretation:

3 Problem 3: Paired t-Test

A physical therapist measures knee flexibility (degrees of extension) in 10 patients before and after a 6-week treatment program:

Patient	Before	After
1	115	125
2	110	120
3	120	128
4	105	115
5	118	130
6	112	122
7	108	118
8	122	135
9	114	124
10	116	126

Part A: Calculate the differences (After - Before) for each patient.

Differences: ____

Part B: Calculate the mean and standard deviation of the differences.

$\bar{d} =$ __________ $s_d =$ __________

Part C: State the hypotheses for testing if flexibility improves after treatment.

H_0: ________________________________

H_A: ________________________________

Part D: Perform a paired t-test in R.

R code:

Test statistic: $t =$ __________ **p-value:** __________

Part E: What is your conclusion? Does the treatment significantly improve flexibility?

Part F: Why would it be wrong to use an independent two-sample t-test for this data? Explain.

4 Problem 4: One-Way ANOVA

A researcher tests the effect of four different exercise programs on weight loss (kg) over 3 months. She randomly assigns participants to one of four programs:

Walking	Swimming	Cycling	Gym
3.2	4.5	5.8	6.2
2.8	4.8	6.1	6.5
3.5	4.2	5.5	5.8
3.0	4.6	6.0	6.3
3.3	4.4	5.7	6.0
Mean: 3.16	Mean: 4.50	Mean: 5.82	Mean: 6.16
SD: 0.27	SD: 0.23	SD: 0.24	SD: 0.27

Part A: State the null and alternative hypotheses for an ANOVA test.

H_0: ______________________________

H_A: ______________________________

Part B: Why is it better to use ANOVA instead of multiple pairwise t-tests?

Part C: Perform a one-way ANOVA in R.

R code to create data frame and run ANOVA:

F-statistic: $F =$ __________

Degrees of freedom: $df_1 =$ ________, $df_2 =$ ________

p-value: __________

Part D: What is your conclusion from the ANOVA? Does it appear that exercise programs differ in effectiveness?

Part E: If the ANOVA is significant, perform Tukey's HSD post-hoc test to determine which programs differ from each other.

R code:

Which pairwise comparisons are significant? (List them)

Part F: Based on your analysis, which exercise program would you recommend for weight loss? Justify your answer.

5 Problem 5: Choosing the Right Test

For each scenario below, identify which statistical test is most appropriate and explain why.

Scenario A: A doctor measures blood glucose in 15 diabetic patients before and after taking a new medication.

Test: ____________________

Reason:

Scenario B: A researcher compares average IQ scores between children who attended preschool (n=25) and children who did not attend preschool (n=30).

Test: ____________________

Reason:

Scenario C: A study compares patient satisfaction scores across five different hospitals.

Test: ____________________

Reason:

Scenario D: An agronomist compares crop yield for two fertilizer brands, using 20 fields. Each field is split in half, with one fertilizer applied to each half.

Test: ____________________

Reason:

Scenario E: A pharmaceutical company wants to determine how many patients to enroll in a trial to detect a 15-point reduction in pain score with 85% power.

Test/Analysis: ____________________

Reason:

Reflection Questions

1. Explain the relationship between Type I error, Type II error, and power. How are they connected?

2. What are the three main factors that affect statistical power, and how does each one influence it?

3. When should you use a paired t-test instead of an independent two-sample t-test? Give an example.

4. What information does an ANOVA F-test give you? What doesn't it tell you?

5. Why is it important to calculate power and sample size *before* conducting a study rather than after?